Student Marks Matrix Analysis (Case Study 42)

Problem Statement

Store marks for students in a nested list and calculate each student's total and average.

Objectives

Practise nested lists, nested loops, sum(), and len().

Assumptions

There are three students and three subjects. Marks are between 0 and 100.

Algorithm

1. Store marks in a nested list. 2. For each student, calculate total and average. 3. Print the results. 4. Find the highest
total.

Flowchart (Text Form)

START → Create marks matrix → Loop through each student row → Calculate total and average → Print → STOP.

Pseudocode

```
a
```

Python Source Code

```
marks = [
    [80, 75, 90],
    [65, 70, 72],
    [92, 88, 95]
]
for i in range(len(marks)):
    total = sum(marks[i])
    average = total / len(marks[i])
    print("Student", i + 1, "total:", total,
          "average:", round(average, 2))
totals = []
for row in marks:
    totals.append(sum(row))
top_index = totals.index(max(totals))
print("Highest total: Student", top_index + 1)
```

Test Case and Expected Output

Given marks in code
Expected: Student 1 total 245 average 81.67; Student 2 total 207 average 69.0; Student 3 total 275 average 91.67; Highest total: Student 3

Conclusion

The program uses a nested list to summarize student marks and find the highest total.